Partitioning of Soil Organic Carbon in Blue Carbon Ecosystems – a quantitative review

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Introduction

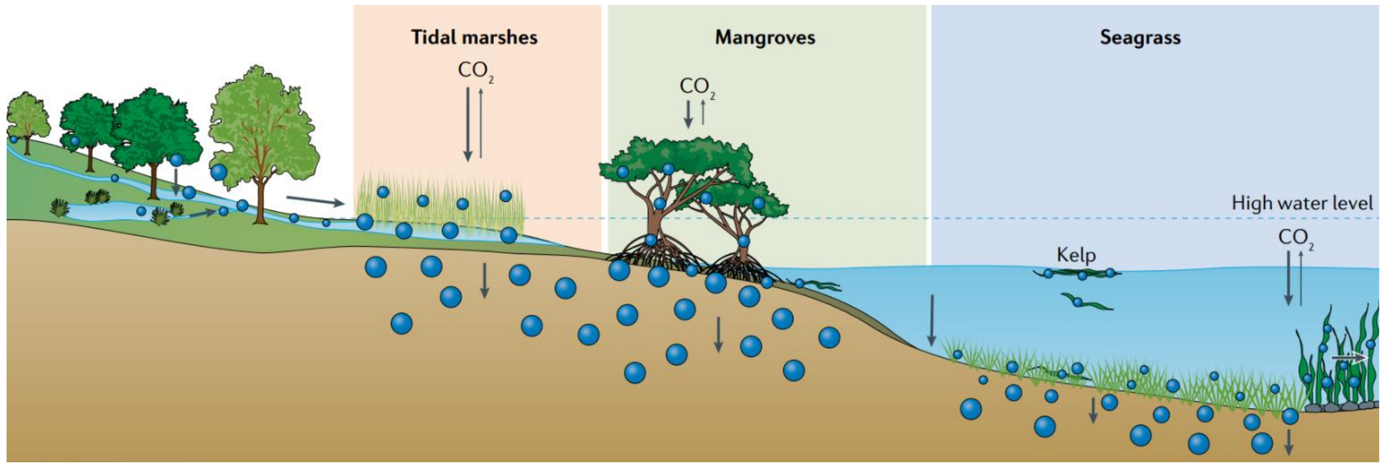


Figure 1. Blue carbon ecosystems (Macreadie and al.[1])

Blue carbon ecosystems (BCEs) sequester carbon and can preserve it for several thousand years in their soils[2]. Yet, little is known about the composition of soil organic carbon (SOC) and particularly its partitioning to the presumably stabilized mineral-associated organic matter (MAOM) pool, and the factors controlling this partitioning.

Research hypothesis:

1/ MAOM will be a major pool across all Blue C ecosystems

2/ Oxygen availability, redox dynamics and salinity might be the major controlling factors of MAOM formation

Methods

- A systematic literature review to identify papers on blue carbon ecosystems reporting mineral-associated organic carbon (MAOC)
- Final search string returned 784 papers (Scopus) & 437 papers (Web of Science). After removing duplicates = 806 papers
- After first screening based on titles and abstracts = 150 articles
- After entire text **screening = 35 articles including 636 measurements**

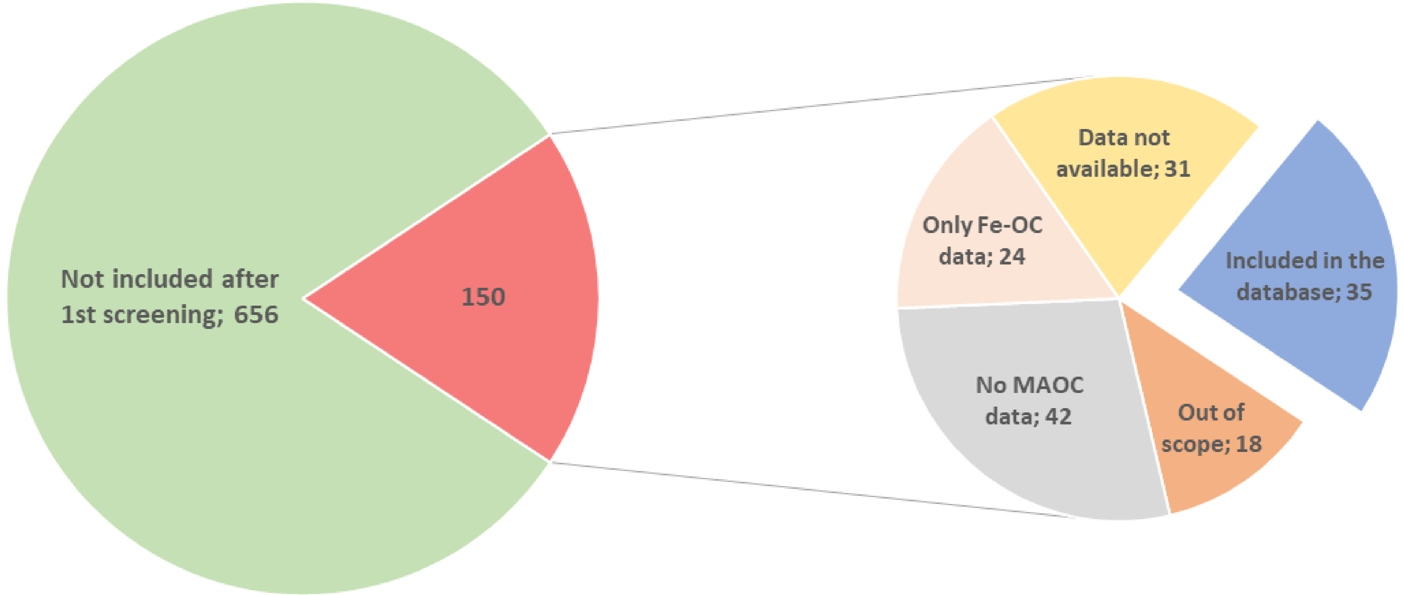
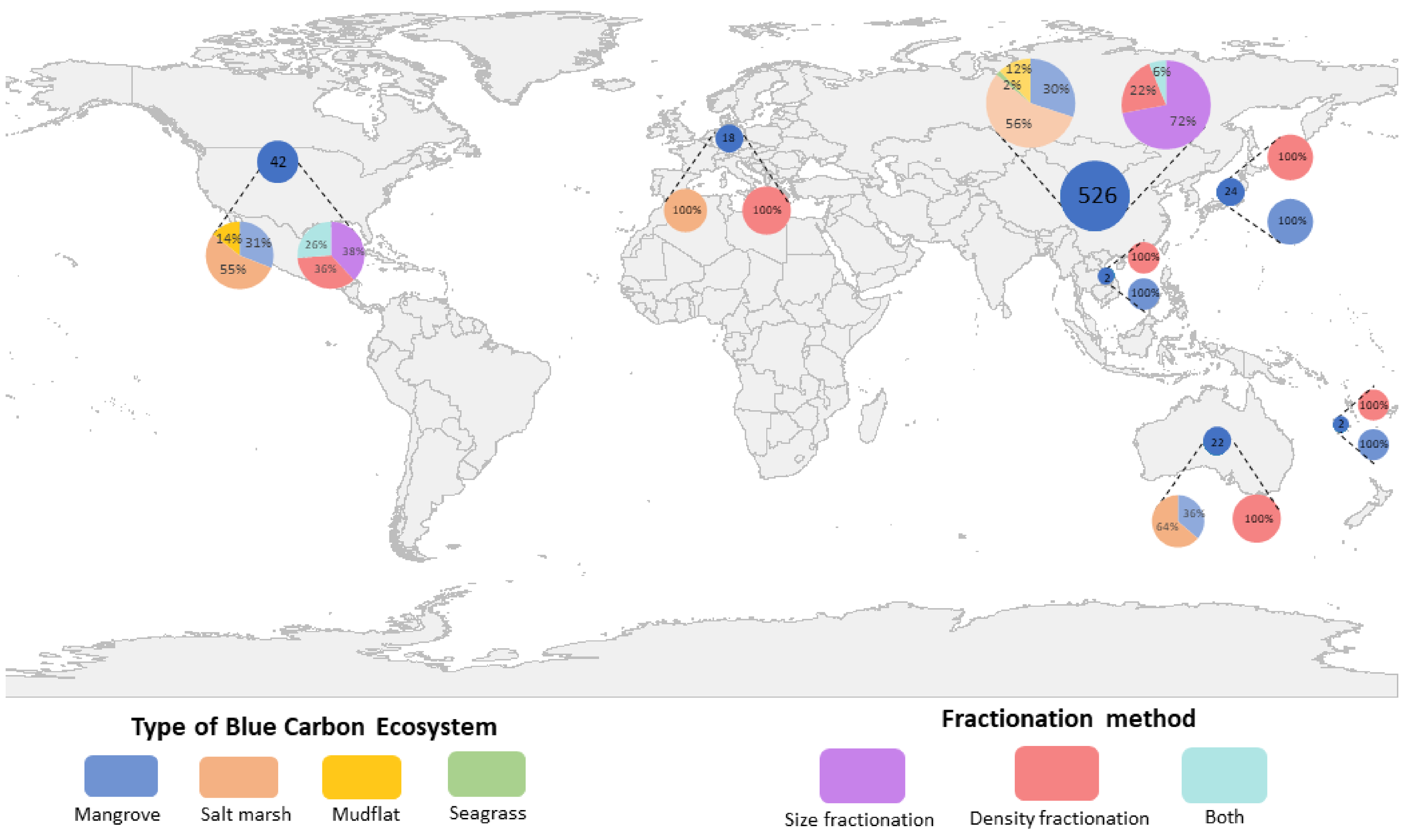
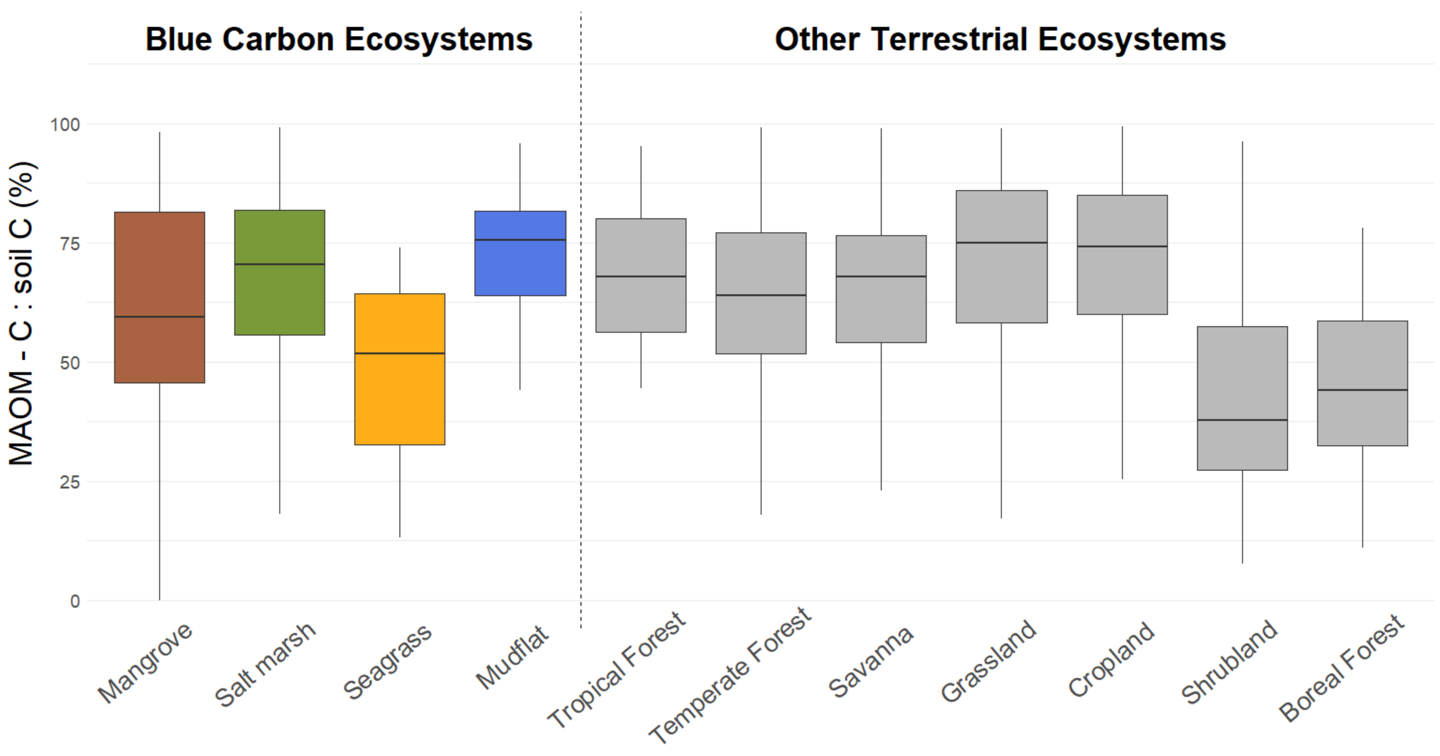


Figure 2. Article selection process

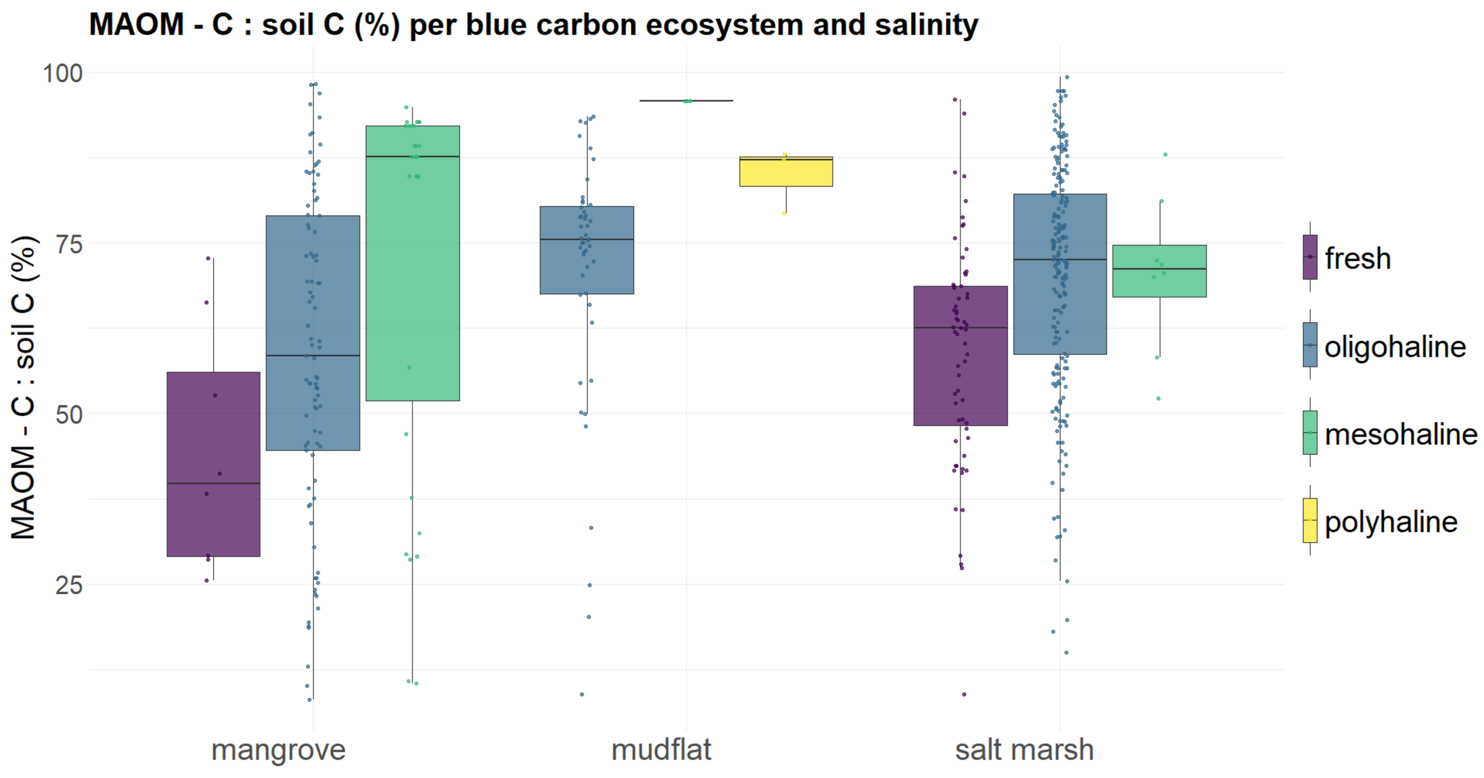
Results and discussion

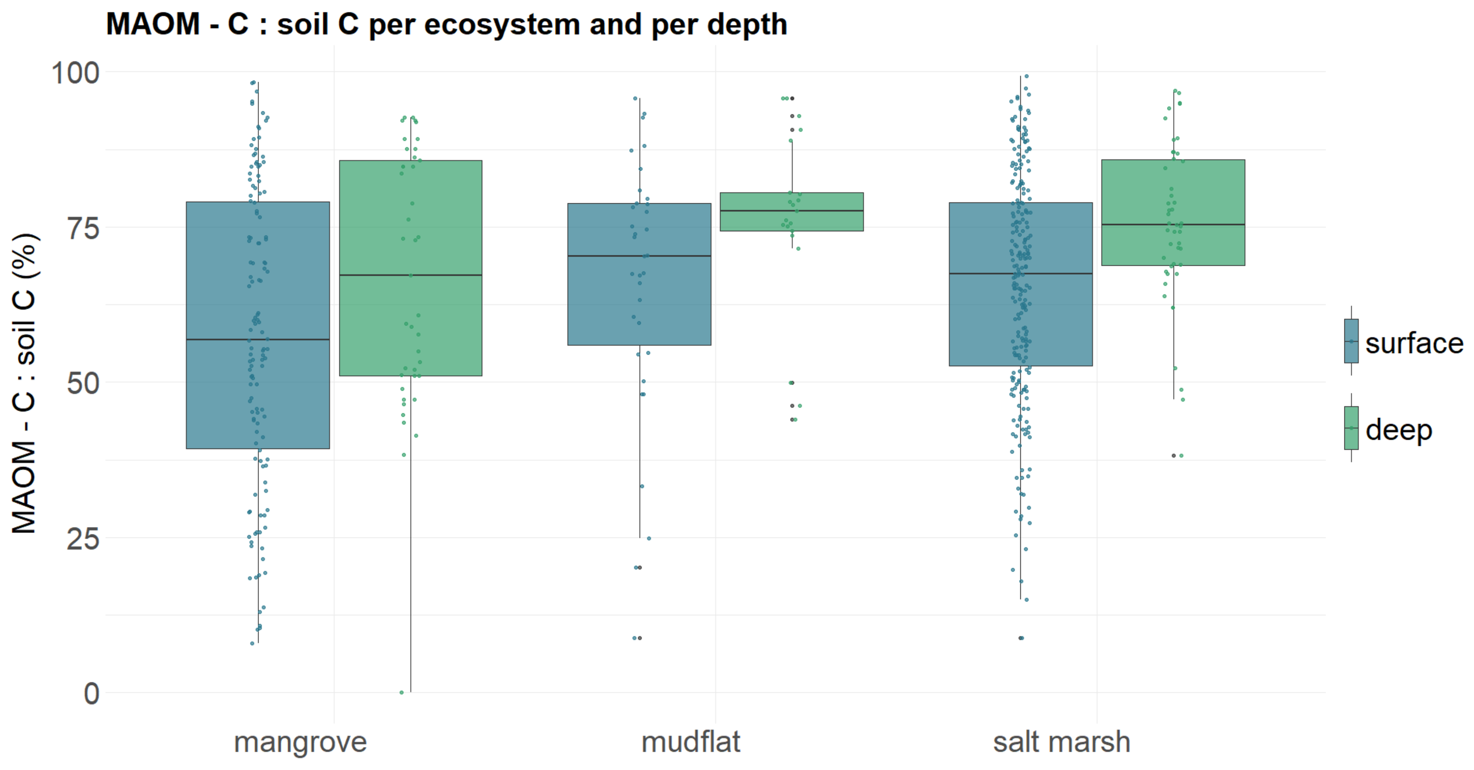


Less than 5% of countries with BCEs are represented, with more than 80% of the measurements in China.



The MAOM pool in BCEs falls within the range observed in terrestrial soils, with the lowest values found in mangroves and the highest in mudflats, although mangrove soils exhibit greater variability.





The MAOC/SOC ratio seems to increase with salinity across all blue carbon ecosystems and seems higher in deep soils (>40cm) than in surface soils (0-20cm). Seagrasses were excluded from the analysis due to insufficient data.

Preliminary findings

- MAOC sequestration in BCEs remains relatively understudied worldwide
- Overrepresentation of studies from China
- Lack of data on MAOC quantification in seagrass meadows ecosystems and most of the data from salt marshes ecosystems
- Further analyses will adopt a statistical approach to assess how environmental parameters (salinity, air temperature, pH, C:N ratio…), and spatial variability (depth gradients and geomorphological settings) affect MAOC storage across BCEs

REFERENCES

1. Macreadie and al., « Blue carbon as a natural climate solution »
2. Iroshaka Gregory Marcelus Cooray and al., « A Review of Properties of Organic Matter Fractions in Soils of Mangrove Wetlands »
3. Sokol and al., « Global distribution, formation and fate of mineral-associated soil organic matter under a changing climate »

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